Daaman Shah

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EDUCATION

Binghamton University New York, USA

M.S. in Computer Science | Artificial Intelligence Track **Dec 2026**

Relevant Coursework: (GPA: 3.97/4.0) - Operating Systems, Systems Programming, Computer Vision, Design and Analysis of Algorithms, Deep Learning, AI, Machine Learning (ML), Design Patterns, Data Mining, Web Programming

TECHNICAL SKILLS

- **Programming Languages**: JavaScript, TypeScript, Python, Java, C++, SQL, HTML5, CSS3
- **Frameworks & Libraries**: React.js, Next.js, Redux Toolkit, Tailwind CSS, Spring Boot, .NET, Flask
- **Backend & APIs**: Node.js, Express.js, RESTful APIs, FastAPI, GraphQL, JWT Auth, OAuth 2.0/OIDC, WebSockets
- **AI/Computer Vision:** Retrieval Augmented Generation, Prompt Engineering, OpenCV, TensorFlow, PyTorch, scikit-learn
- **Databases & Data Tools:** PostgreSQL, MySQL, MongoDB, Redis, Prisma, Firebase Realtime DB
- **Cloud & DevOps:** AWS EC2, Docker, Kubernetes, GCP, Unix/Linux, Terraform, GitHub Actions, CI/CD Pipelines
- **Development Tools:** Git, GitHub, npm, Jira, Figma, Postman, Selenium, GoogleTest, GitHub Copilot

PROFESSIONAL EXPERIENCE

Watson School of Computing - Binghamton University Binghamton, NY

Software Engineer Dec 2025 – Present

- Led Watson Tech Talent, a project marketplace (**Next.js, TypeScript, PostgreSQL**), enabling faculty across departments to recruit students for research and technical work while extending university's IT capacity by routing suitable projects
- Implemented **BM25** ranking to evaluate resumes and application responses against project requirements, helping sponsors identify relevant applicants beyond static profile information
- Reduced manual outreach by **80%** through automated segmented email and in-app notifications and routine developer intervention by approximately **60%** through an admin operations system
- Refactored **80% of the frontend** into modular, reusable components, resolved UI defects, and passed **WCAG 2.1 accessibility checks**
- Implemented **role-based access control (RBAC)**, standardized **PostgreSQL migration** workflows and schema consistency rules, and automated server deployments through **GitHub Actions CI/CD**

Konze Pvt Ltd Ahmedabad, India

Software Engineer Intern Jun 2024 – Dec 2024

- Contributed to an **OCR-based document processing pipeline** for KonDesk, a cloud CRM for education and immigration consultants, extracting text from client documents and cross-referencing against eligibility checklists
- Reproduced and resolved defects across **React state, Node.js/Express APIs, and third-party integrations**, validating fixes through team testing in Git- and Jira-managed Agile sprints

PROJECTS

CoursePilot AI - AI Powered Learning System

- Developed an AI-powered academic learning system that turns instructor-provided course content into guided study, tutoring, and practice experiences using **React, Node.js, PostgreSQL, and Prisma**
- Implemented a course-grounded **RAG tutor** that retrieves instructor approved content to generate contextual explanations and progressive hints
- Built asynchronous document processing workflows with background queues and **Redis caching**, separating ingestion from interactive tutor requests, standardized application environments with **Docker** for consistent setup and deployment

Schrödinger's Human - Human or AI?

- Built an anonymous human or AI guessing chat with Node.js, TypeScript, Socket.IO, Redis, and PostgreSQL, achieved **sub-140 ms p95** message delivery across **50 concurrent conversations and 100 simulated clients**
- Developed a **Python/FastAPI** service for context-aware AI participants using conversation history and persona-based prompting, coordinated asynchronous generation with durable leases and stale-response protection

GameVault - Game Search & Recommendation Platform

- Designed a game discovery interface with Next.js, TypeScript, and Redux Toolkit, combining RAWG catalog search with **OpenAI GPT-5.6-Luna** for AI-powered recommendations
- Improved **search performance by 31%** by implementing **RTK Query caching, debounced API requests, responsive image optimization, and lazy loading**, reducing redundant network calls and page-transfer overhead